

ABSTRACT ALGEBRA 1 - HOMEWORK 4- M. VICARÍA

Dear student please submit the complete solution to each of the following problems in Gradescope in a PDF file. Please write down your name in each page and specify clearly the problem that you stated. For your study purposes, I included a classification of the problems and their difficulties:

- Warm-up: these are require that you understand some definition.
- *: they indicate some medium level of difficulty. You should spend longer in these problems.
- ** indicates that the problem is challenging, they will require more time than the other problems.

The complexity from ** and * is the level we would like you to master in this course. Questions highlighted with **OPTIONAL** are not necessary to get full credit for the homework, but I highly encourage you to solve them for your learning process.

In each of the solutions you **SHOULD JUSTIFY COMPLETELY YOUR ANSWER** with a proof or a counter-example. No credit is granted to solutions without justification.

1. MANDATORY PROBLEMS

- (1) **Problem 1***: Classify all the subgroups of $(\mathbb{Z}_{36}, +, 0)$.

- (2) **Problem 2**:** Show that $(\mathbb{R}, +, 0)$ is not finitely generated.

Note that when you consider $\mathbb{R}$ as a vector space it has dimension one, so when you forget the scalar multiplication you are actually losing a lot of structure.

- (3) **Problem 3*:** Let $(G, \star)$ and $(H, \cdot)$ be two groups. Assume $(G, \star)$ is generated by $g_1, \dots, g_n \in G$. Prove that if ϕ and ψ are two homomorphisms from G to H then:

If for all $i \leq n$ we have $\phi(g_i) = \psi(g_i)$ then $\phi = \psi$.

This proves the important fact that a homomorphism is completely determined by what it does to a generating set.